THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 6: **We Distribute, Yet Things Multiply** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of 98×12 is: 1

(a) 1176 **(b)** 1164 **(c)** 1188 **(d)** 1186

2. The value of 97×14 is: 1

(a) 1368 **(b)** 1372 **(c)** 1344 **(d)** 1358

3. The value of 52^2 is: 1

(a) 2714 **(b)** 2704 **(c)** 2756 **(d)** 2652

4. The value of $53^2 - 47^2$ is: 1

(a) 5018 **(b)** 600 **(c)** 36 **(d)** 10000

5. The value of $64^2 - 36^2$ is: 1

(a) 2800 **(b)** 5392 **(c)** 784 **(d)** 10000

6. If $a = 6$ and $b = 1$, then $(a + b)^2$ equals: 1

(a) 25 **(b)** 37 **(c)** 43 **(d)** 49

7. The property $a \times (b + c) = a \times b + a \times c$ is called: 1

(a) commutative property **(b)** associative property
(c) closure property **(d)** distributive property

8. The product of any number and 0 is: 1

(a) 1 **(b)** the number itself
(c) 0 **(d)** undefined

9. The expression $a^2 + 2ab + b^2$ is the expansion of: 1

(a) $a^2 - b^2$ **(b)** $(a + b)^2$ **(c)** $(a + b)(a - b)$ **(d)** $(a - b)^2$

10. The area of a square of side $(x + 2)$ is: 1

(a) $4x + 4$ **(b)** $x^2 + 4x + 4$ **(c)** $x^2 + 4$ **(d)** $x^2 + 2x + 4$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State the distributive property of multiplication over addition. 2

12. Find 98×15 using the distributive property. 2

13. Expand $(x + 6)^2$. 2

14. Expand $(2a - 3)^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Evaluate $41^2 - 39^2$ without calculating each square. 3

16. Evaluate $68^2 - 32^2$ without calculating each square. 3

17. Evaluate $97^2 - 3^2$ without calculating each square. 3

OR

Evaluate $205^2 - 195^2$ without calculating each square. 3

18. Evaluate $73^2 - 27^2$ without calculating each square. 3

OR

Evaluate $111^2 - 89^2$ without calculating each square. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Derive $(a + b)^2$ using an area model.
(ii) Evaluate 73^2 using it. 5

OR

(i) State the identity for $(a - b)^2$.
(ii) Evaluate 196^2 using $(200 - 4)^2$. 5

20. (i) Prove $(a + b)(a - b) = a^2 - b^2$.
(ii) Compute 52×48 and 103×97 using it. 5

OR

Case study: A square courtyard of side $(n + 3)$ m has a square flower bed of side n m at one corner. 5
(i) Write the area of the courtyard.
(ii) Write the area of the remaining part.
(iii) Find it for $n = 7$.

END OF QUESTION PAPER — Total: 40 Marks

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